out compromising safety or performance. The Avishkar team competed with their Pod 5.0 at the European Hyperloop Week 2023 in Delft, the Netherlands, which is when they knew they were headed in the right direction. Avishkar ranked in the Global Top-5 in the Electrical, Traction and Complete Pod categories. The team is now busy preparing for this year's iteration of the competition, which will happen in July. IIT Madras also organised a Global Hyperloop Competition of its own in February, which saw the participation of 10 Hyperloop teams from across the world.

One of the highlights of the Avishkar project is the prototype concrete tube. However, "concrete is porous. So, preserving a vacuum will be a challenge. In a year or two, we can complete the tube," claims Prem. Their 'booster-cruiser system' employs two kinds of motors. The pod is driven by a single motor. Another motor further powers the moving pod at a constant velocity and is very energy-efficient. Cabin systems onboard—such as life support, humidity control and thermal management—are now in the testing stage, using procedures similar to those employed in space flight. The pod uses a mission control-style GUI (graphical user interface) network that gathers and displays real-time data, allowing it to monitor and even perform diagnostics on the go.

IS IT A PIPEDREAM?

The first-ever manned Hyperloop test was held in 2020 by Virgin Hyperloop, once the poster child for the movement, reaching a top speed of 172 km per hour. Since then, a string of private efforts have stumbled on commercial roadblocks, shifting priorities and scaling issues. "The dream is still alive," they say, but not without glitches. Though Musk has proposed that journeys—say, from Los Angeles to San Francisco or New York to Washington, DC—could take under 30 minutes, test velocities of unmanned pods under practical conditions have hovered around 387 km per hour due to safety, engineering, and scaling limitations.

No wonder, the road to commercialisation remains hazy worldwide. Some research firms don't expect Hyperloop to be commercially viable before 2040. While developers claim that artificial intelligence is accelerating progress, Indian policymakers are keeping room for some healthy scepticism. Vaishnaw himself has called the technology "nascent". Fundamental problems

RACING INTO THE FUTURE

India is accelerating its Hyperloop ambitions with Avishkar, an IIT Madras student-led initiative to build the world's longest test track and take the lead in next-gen transport innovation. This is what the project entails

THE PROJECT: AVISHKAR HYPERLOOP

GOAL
To build the world's longest hyperloop test track (40-50 km)

SUPPORT
Rs 30+ crore from Modi government since 2022

FIRST-USE CASE
High-speed cargo movement via Indian Railways

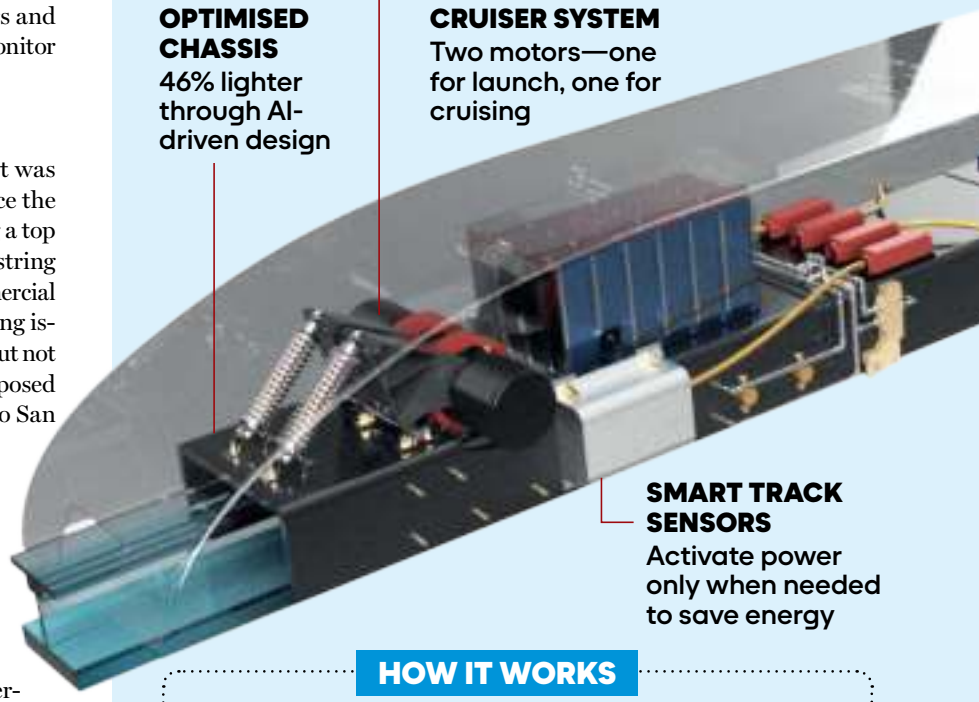
TARGET SPEED

1,200 KMPH
Faster than a commercial jet

CORE TECHNOLOGIES

TOPOLOGY-OPTIMISED CHASSIS
46% lighter through AI-driven design

BOOSTER-CRUISER SYSTEM
Two motors—one for launch, one for cruising



SMART TRACK SENSORS
Activate power only when needed to save energy

HOW IT WORKS

- With the air removed, the vacuum tube eliminates drag
- Electromagnetic levitation lifts the pod off the surface
- The linear electric motor provides thrust
- A near-frictionless glide at breathtaking velocity